Microsoft Intune + Windows Device Management Implementation Plan

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Version: 1.0 I **Date:** 25 June 2025

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1. Executive Summary

This implementation plan expands on the original high-level phases and introduces detailed technical execution steps, architectural context, dependencies, third-party components, and a Microsoft Project-style task breakdown. It addresses client expectations around traceability, reproducibility, and technical decision-making for OT/field environments.

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2. Architectural Diagrams & Design

2.1 Identity Integration Flow (Hybrid Join, AAD Sync, MFA)

This flow ensures that devices join Azure AD while retaining their connection to on-premises Active Directory, enabling centralized management and conditional access enforcement. Key components include: - Azure AD Connect: Syncs users, groups, and device objects from on-prem AD to Entra ID. - Hybrid Azure AD Join (HAADJ): Devices join both on-prem and Azure AD using SCP and Group Policy. - MFA (Duo Security or Entra MFA): Applied at login or conditional access levels, ensuring strong authentication across environments.

Process Flow:

1. Device connects to domain controller (GPO sets SCP)

2. Azure AD Connect syncs object to cloud

3. Device uses scheduled task or Autopilot to register with Entra ID

4. MFA triggered during login or resource access (via Conditional Access)

2.2 Device Compliance Enforcement Pipeline

This architectural layer enforces device health, configuration posture, and access controls.

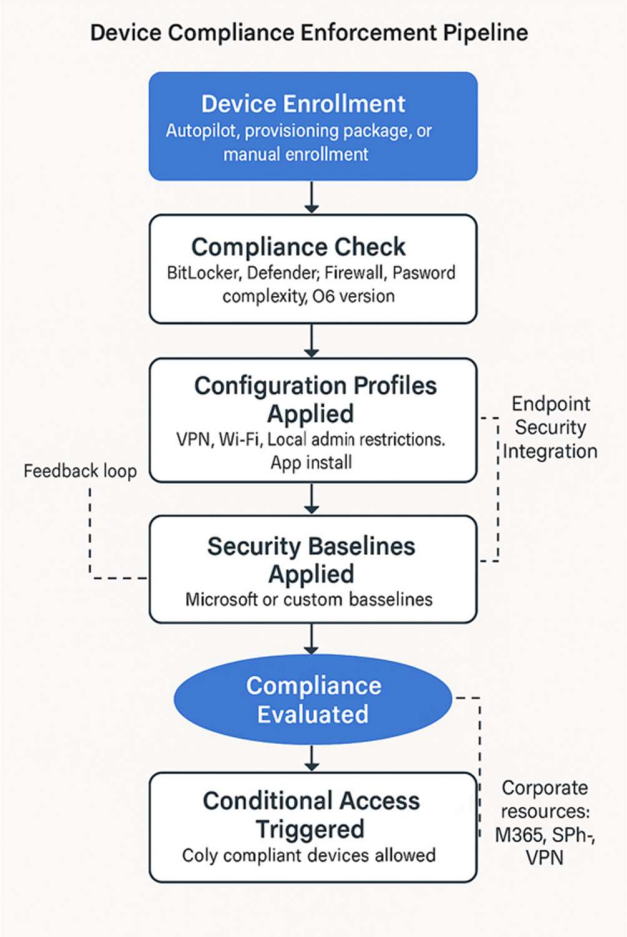
Core Steps:

1. Device Enrollment: Devices enroll via Autopilot, provisioning package, or manual enrollment
2. Compliance Check: BitLocker, Defender, Firewall, Password complexity, OS version

3. Configuration Profiles Applied: VPN, Wi-Fi, Local admin restrictions, App install

4. Security Baselines Applied: Microsoft or custom baselines pushed

5. Compliance Evaluated: Device marked as compliant/non-compliant 6. Conditional Access Triggered: Only compliant devices allowed access to corporate resources (M365, SharePoint, VPN)



2.3 5G/SIM Restricted Connectivity Model

Designed to restrict access to specific cloud endpoints when using mobile broadband (e.g., embedded SIM in field laptops).

Components: Device has embedded or external 5G SIM

- Wi-Fi & Ethernet interfaces disabled via script

- DNS filtering via Cisco Umbrella or Zscaler limits outbound traffic

- APN profiles pre-defined (OEM-dependent)

- VPN enforced to connect to corporate network

Challenges: Intune does not natively manage SIM policies deeply - Requires scheduled script enforcement - Modem vendor tools or registry-level CSP may be required

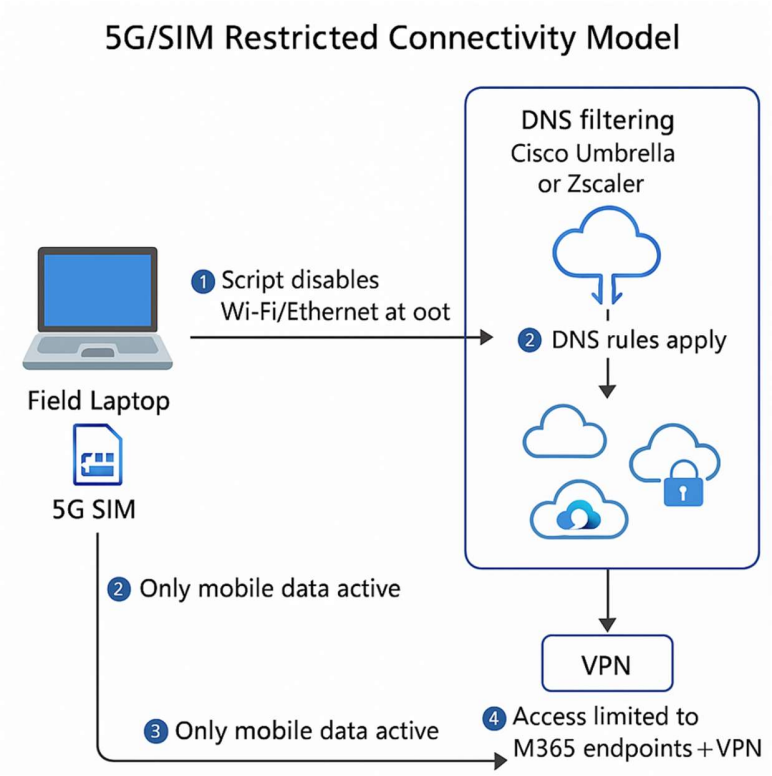
Flow:

1. Script disables Wi-Fi/Ethernet at boot

2. Only mobile data active

3. DNS rules apply

4. Access limited to M365 endpoints + VPN



2.4 USB Control Architecture (Defender E5 vs. 3rd Party)

This architecture secures USB port usage to prevent data exfiltration and unauthorized tool usage.

Scenario A: Microsoft Defender for Endpoint (E5 license) - Device Control Policy applied via Intune - USB devices filtered by Vendor ID/Product ID (VID/PID) - Allowed devices whitelisted (e.g., USB-Serial converters) - Logging and alerting integrated with MDE console

Scenario B: 3rd Party Tool (e.g., DeviceLock, Safend) - Agent deployed via Intune (Win32 app) - Advanced control over USB, Bluetooth, Wi-Fi - Central management console (may require server or cloud console) - More flexible for OT/legacy driver scenarios

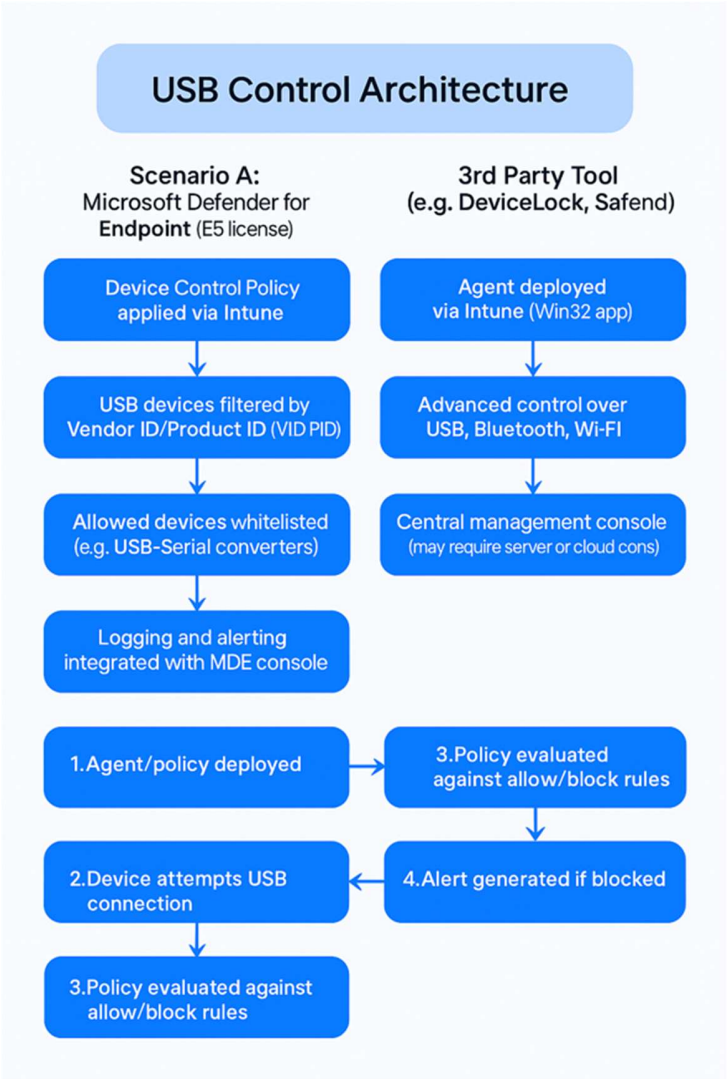
Flow:

1. Agent/policy deployed

2. Device attempts USB connection

3. Policy evaluated against allow/block rules

4. Alert generated if blocked



2.5 Pilot vs. Production Device Enrollment Path

Defines how devices are onboarded for initial testing vs. full rollout.

Pilot Path: Limited to 2–3 laptops

- Manual enrollment or provisioning package

- Assigned to “Pilot Devices” group

- All policies and use cases tested

- Device logs, screenshots, and config captured

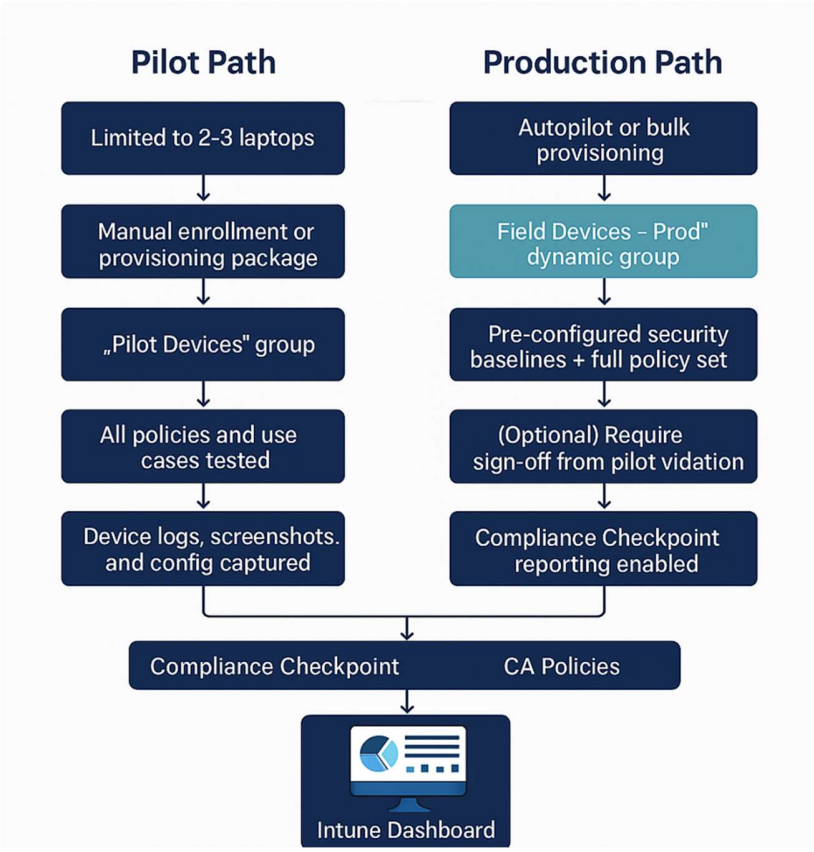
Production Path: Devices onboarded via Autopilot or bulk provisioning

- Assigned to “Field Devices - Prod” dynamic group

- Pre-configured security baselines + full policy set

- Logging and compliance reporting enabled

- Optional: Require sign-off from pilot validation



3. Detailed Phase and Task Breakdown

The following table outlines all key project phases with task-level detail, presented in a Microsoft Project-style matrix format. Each task includes assigned team, tools/software, timeline estimate, dependencies, and expected deliverables.

Phase 1: Tenant Preparation & Identity Integration

Task ID	Task Name	Description	Assigned Team	Tools	Dependencies	Deliverables
1.1	Domain Setup	Verify and register custom domain in Microsoft 365	Tenant Admin	M365 Admin Center	Global Admin Rights	Verified DNS status
1.2	Licensing	Assign Intune, Defender, AAD Premium licenses	Tenant Admin	Admin Center	License Availability	Licensing matrix
1.3	DNS Record Validation	Add TXT and MX for domain verification	Network Engineer	Registrar portal	Domain Access	DNS test pass
1.4	AAD Connect Install	Install AD Connect in staging mode	AD Admin	AAD Connect, RDP	Jump server, AD access	Configured AD Connect
1.5	GPO Configuration	GPO for SCP, Hybrid Join, MFA	GPO Admin	GPMC	Domain Controller Access	Linked GPO objects
1.6	Firewall Ports	Open required ports (443, 80, 389, etc.)	Security	Firewall console	Network Team Access	Firewall rules applied
1.7	Hybrid Join Test	Validate device registration to Entra ID	Field Tech	dsregcmd, Event Viewer	AAD Connect sync	Device status proof
1.8	MFA Compatibility	Test Duo MFA with hybrid join	Security	Duo Admin, AD	Working MFA setup	Duo login log

Phase 2: Intune Configuration & Policy Baselines

Task ID	Task Name	Description	Assigned Team	Tools	Dependencies	Deliverables
2.1	MDM Authority	Set Intune as primary MDM	Intune Admin	Endpoint Manager	Admin Access	MDM flag enabled
2.2	Compliance Policies	Create policies for BitLocker, AV, firewall	Security Lead	Intune Portal	License assignment	JSON exports
2.3	Config Profiles	Create profiles (Wi-Fi, VPN, kiosk)	Intune Admin	Intune	Network info, Device specs	Profile list
2.4	Security Baselines	Deploy Microsoft security baselines	Security	Baseline templates	Device groups	Assigned baseline snapshot
2.5	Group Setup	Create pilot/production groups	Intune Admin	AAD Groups	Naming convention	Group rules + scope tags
2.6	Profile Targeting	Apply profiles and test push	Intune Admin	Intune console	Group readiness	Deployment proof

Phase 3: Secure Use Case Implementation

Task ID	Use Case	Description	Assigned Team	Tools	Dependencies	Deliverables
3.1	5G SIM Restriction	Disable Wi-Fi & Ethernet, apply DNS filtering	Intune + SecOps	PowerShell, Cisco Umbrella	SIM-enabled device	DNS log, connectivity test
3.2	Disable Wi-Fi	Script + recurring task	Intune Admin	Scheduled Task, PowerShell	Device model	Device snapshot

		to block adapter				
3.3	USB Whitelist	VID/PID allow list config	Security + MDE	Defender E5 / DeviceLock	Device control license	Allow/block logs
3.4	USB-Serial Support	Package pre-approved drivers	Packaging Engineer	Intune + Win32	Driver access	Deployment proof
3.5	Offline Access	Configure cached login, enforce policies offline	GPO + Intune	GPMC, local device tests	Intune + GPO active	Policy validation screenshots
3.6	Patch Tool Validation	Enable patch access via VPN/SICE	Intune + Infra	DNS/Firewall rules	SICE access	Patch tool test logs

Phase 4: Pilot Enrollment & Testing

Task ID	Task Name	Description	Assigned Team	Tools	Dependencies	Deliverables
4.1	Select Pilot Devices	Choose 2–3 field-ready laptops	Infra + PM	Asset Inventory	Device list	Device IDs listed
4.2	Manual Enrollment	Use provisioning package for test devices	Field Tech	Windows ICD, Intune	Enrollment profile	Enrolled device list
4.3	Policy Validation	Confirm policy/app/config applied	Field Tech	Intune portal, screenshots	Profile readiness	Compliance dashboard
4.4	Offline Scenario Test	Duo MFA, offline login, USB tools	Security + Infra	Manual test cases	MFA + offline config	Screenshot & results sheet
4.5	Logging & Troubleshooting	Capture logs for failed items	All Teams	Event Viewer, Intune	Completed tests	Log package

Phase 5: Full Deployment

Task ID	Task Name	Description	Assigned Team	Tools	Duration	Dependencies	Deliverables
5.1	Device Rollout	Enroll all field laptops	Field Support	Autopilot / Manual	2 days	Device prep	Enrolled devices list
5.2	Compliance Tracking	Monitor policy success via Intune	Intune Admin	Compliance reports	1 day	Policy push	Report snapshot

4. Use Case Deep Dive

Each case below is structured to explain the reasoning, exact steps, tooling requirements, and fallback options, ensuring field-readiness and audit alignment.

Use Case 1: Restrict Internet Access via 5G SIM

Background & Justification: Field devices need secure, controlled internet access limited to Microsoft and VPN endpoints while connected via cellular (5G SIM). Preventing Wi-Fi/Ethernet use reduces attack surfaces.

Implementation Plan:

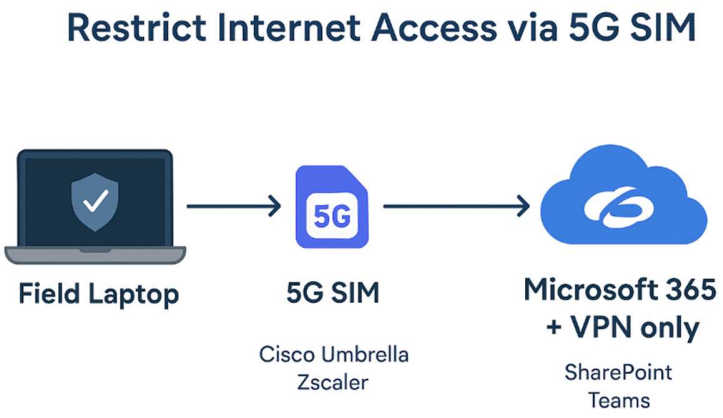
- Disable Ethernet and Wi-Fi adapters at OS level using PowerShell.
- Enforce recurring scheduled tasks to block adapter re-enablement.
- Apply DNS filtering (Cisco Umbrella/Zscaler) to allow only M365, VPN, ShareFile domains.
- Preconfigure APN profile if OEM support exists.

Tools: PowerShell, Cisco Umbrella, Scheduled Tasks, Intune Device Config

Limitations/3rd Party Needs: OEM driver or APN tool may be needed for complete enforcement. - DNS filtering requires Umbrella or Zscaler license.

Testing Plan: nslookup to non-allowed domain: FAIL - Access to SharePoint/Teams: PASS - Device reboots and retains blocked interfaces

Rollback Strategy: Remove scheduled task - Restore default adapter state via Intune script



Use Case 2: Disable Wi-Fi Adapter

Background & Justification: Prevent users from connecting to untrusted networks in field environments. Disabling Wi-Fi at device level ensures secure traffic routing only through approved SIM or VPN tunnels.

Implementation Plan: Use PowerShell to disable adapter - Apply Intune Device Restriction policy to remove Wi-Fi access in UI - Enforce recurring script using scheduled task

Tools: PowerShell, Intune, Task Scheduler

Limitations: Can be overridden with local admin access if not locked down - Adapter-level block is not persistent on all models without scripting

Testing Plan: Validate adapter shows disabled in Device Manager - User cannot re-enable via Control Panel

Rollback Strategy: Remove scheduled task - Deploy script to re-enable adapter via Intune

Use Case 3: Allow USB-Based IP Configuration Tools (Whitelist USB Devices)

Background & Justification: OT engineers require USB-based IP config tools to set up networked hardware. Security posture demands tight control on USB ports, allowing only approved tools.

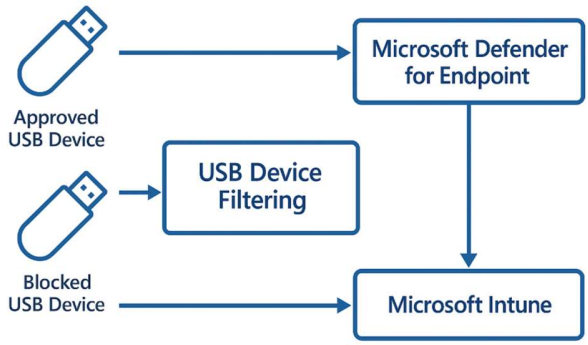
Implementation Plan: Enable Device Control in Microsoft Defender (E5) or use DeviceLock - Collect VID/PID for approved tools - Apply Intune Defender profile or 3rd-party policy

Tools: Microsoft Defender for Endpoint, DeviceLock, Intune Win32 App

Limitations/3rd Party: VID/PID enforcement requires Defender E5 or compatible 3rd-party USB control software

Testing Plan: Plug in approved USB device → Should work - Plug in mass storage → Should be blocked - Log generation in MDE portal

Rollback Strategy: Remove USB policy from Intune - Revert Defender settings to default



Use Case 4: Support Serial Communication via USB-to-Serial

Background & Justification: Field engineers often use USB-to-Serial converters to interface with legacy or industrial equipment.

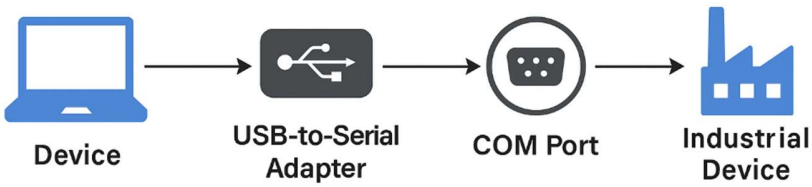
Implementation Plan: Identify approved USB-serial adapters and drivers - Create Win32 Intune package to install drivers - Whitelist device using Defender or GPO policy

Tools: Intune Win32 Packaging, Defender, Custom Driver Staging

Limitations: Some drivers may not install silently or require user confirmation - Defender may block unsigned drivers

Testing Plan: Plug device and check COM port creation - Attempt serial console access - Validate repeatable behavior across reboots

Rollback Strategy: Remove Win32 package from affected devices - Manually uninstall driver if needed



Use Case 5: Secure Offline Access

Background & Justification: Field users must be able to log in and work offline in remote areas, while still maintaining policy compliance and data protection.

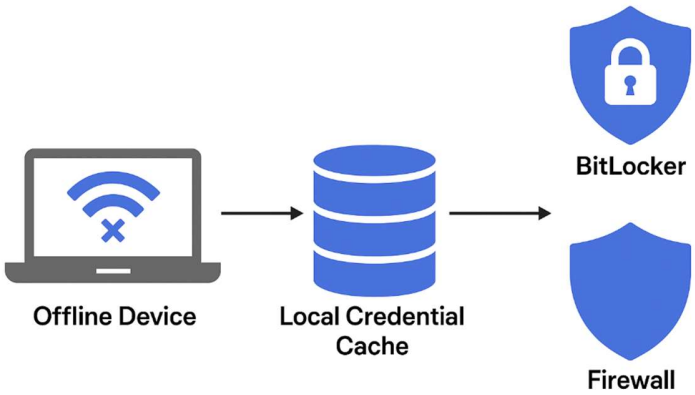
Implementation Plan: Enable cached login (GPO + Intune) - Ensure BitLocker, Defender, Firewall are enforced without internet - Validate policy enforcement during offline boot scenario

Tools: Group Policy, Intune Configuration Profiles

Limitations: Some policies (Defender updates, app install) won't apply without connectivity

Testing Plan: Disconnect internet and reboot - Verify user login success and local policy enforcement - Confirm Defender and firewall state via local tools

Rollback Strategy: Revert GPO settings - Allow full connectivity to re-sync with Intune



Secure Offline Access

Use Case 6: SICE (MSP) Patch Access Validation

Background & Justification: Ensure patching tool (SICE) can reach devices over approved remote session or VPN channel, even in restricted environments.

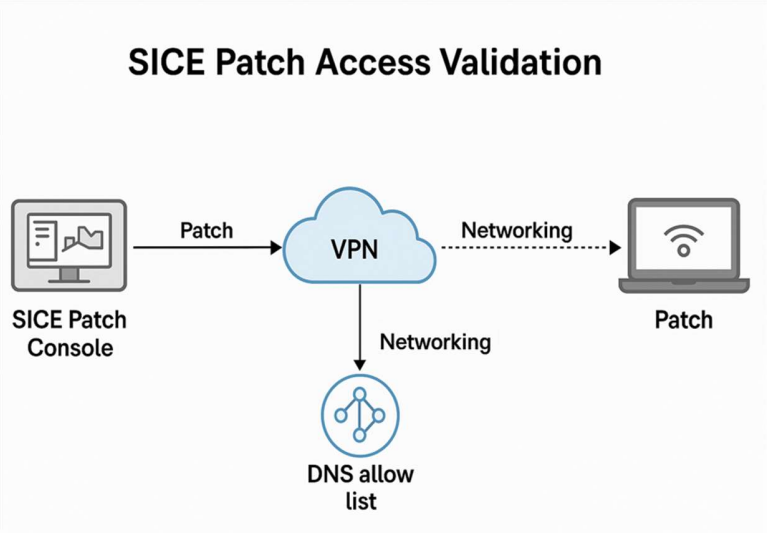
Implementation Plan: Validate VPN profile allows communication to SICE - Apply DNS rules to permit patch tool endpoints - Confirm patch console visibility and job push

Tools: VPN client, DNS filtering service, Patch Console

Limitations: Over-restrictive DNS filter may block update sites - VPN client may require re-auth post-reboot

Testing Plan: Run patch job and check delivery - Simulate DNS block and retry - Record latency and success/failure logs

Rollback Strategy: Remove SICE package and policy - Switch to fallback patch process (e.g., Intune LOB updates)



5. Policy and Configuration Catalog

This section outlines the catalog of compliance policies, configuration profiles, baselines, and device groups used throughout the implementation. All configurations are exportable in JSON or XML format

5.1 Compliance Policies

Policy Name	Description	Enforcement Area	Platform	Notes
BitLocker Compliance	Ensures full disk encryption with TPM	Endpoint Security	Windows 10/11	Requires TPM + Recovery key escrow
Antivirus Enforcement	Windows Defender AV must be enabled and updated	Endpoint Security	Windows	Custom alert threshold set
Firewall Enforcement	Firewall must be active across all profiles	Endpoint Security	Windows	All network types enforced

Password Policy	Enforces strong passwords (8+ chars, complexity)	Device Compliance	Windows	Warning sent at 5 days expiry
Minimum OS Version	Requires Windows 10 21H2 or later	Device Compliance	Windows	Blocks outdated versions

5.2 Configuration Profiles

Profile Name	Description	Key Settings	Delivery Method
VPN Config	Auto-connect VPN for field devices	Always-On, Split Tunnel	Device Profile
Wi-Fi Setup	Predefined secure SSID for in-office use	WPA2 Enterprise	Device Profile
Shared/Kiosk Mode	Assigned access to restrict public interaction	Auto-login, App Lockdown	Custom CSP + Intune
Local Admin Block	Removes user from local admin group	Account Protection	Intune Settings Catalog
Lock Screen Enforcement	Custom lock screen message + timeout	Legal Notice + 1-min idle	Device Profile

5.3 Security Baselines

Baseline Name	Description	Based On	Key Settings
Microsoft Security Baseline	Recommended baseline for Windows 10/11	Microsoft	ASR, SmartScreen, Credential Guard
Legacy Protocol Block	Disable SMBv1, NTLM fallback	Custom Baseline	GPO + CSP hybrid model
Custom OT Baseline	Hardened config for field use	Internal Profile	USB lockdown, offline login control

5.4 Device Groups and Dynamic Membership Rules

Group Name	Type	Membership Rule	Targeted Policies
Pilot Devices	Dynamic	(device.deviceName - contains “PILOT”)	All test policies
Field Devices - Prod	Dynamic	(device.extensionAttribute1 - eq “Field”)	Compliance, Config, Security Baseline
Admin Devices	Static	Manual Add	Fewer restrictions, full access
USB Approved Devices	Static	Manual Add	USB policy exceptions

6. GPO and AD Configuration

This phase is critical to validate all design assumptions, policy behavior, and technical configurations in a controlled pilot environment. Before moving to full rollout, 2–3 laptops will be enrolled and tested end-to-end to ensure system stability, enforceability of restrictions, and compatibility with operational workflows (including offline scenarios and third-party patch access).

6.1 Enroll 2–3 Pilot Devices

- Assign pilot users and register their devices in Azure AD (hybrid join).
- Enroll devices into Microsoft Intune via manual or GPO-assisted onboarding.
- Confirm license assignments and MDM scope application.

6.2 Validate Hybrid Join + Intune Enrollment

- Ensure devices appear in both **Azure AD** and Intune portal.
- Confirm device state: “Hybrid Azure AD joined” and “Compliant”.
- Review logs from Azure AD Connect and MDM enrollment.

6.3 Verify All Policies Apply Correctly

- Compliance policy: BitLocker, Defender AV, Firewall, password/PIN rules.
- Configuration profiles: Wi-Fi disabled, VPN applied, kiosk/shared device mode enforced.
- USB device policies: Ensure approved USB camera tools work; unapproved devices blocked.
- Conditional Access (if applicable): Confirm device access is restricted unless compliant.

6.4 Simulated Field Use

- Disconnect from internet → confirm offline login works.
- Launch and operate USB-based camera, UPS, or serial field tools.
- Attempt to connect to internet via Wi-Fi or Ethernet to ensure blocked.
- Verify internet access is routed only through built-in 5G SIM (via APN if configured).
- Validate policy enforcement during offline usage (policy cache remains in effect).

6.5 Validate SICE MSP Access and Patching Tools

- Confirm remote access from SICE jump host, VPN, or Bastion if used.
- Ensure patching or update push tools function correctly on test device.
- Monitor firewall, DNS logs (e.g., via Cisco Umbrella) to verify allowlisting success.
- Capture update logs from Intune or Endpoint Analytics for patch status confirmation.

6.6 Deliverables

- Pilot testing report with screenshots, logs, and outcomes
- Issue/exception tracker for any deviation from expected results
- Validation checklist signed off by internal stakeholders
- Updated recommendations before moving to Phase 5 rollout

7. Pilot Plan and Testing Playbook

This section defines the structure and operational flow for validating the solution through a controlled pilot rollout of 2–3 representative field laptops. Testing covers identity, compliance, connectivity, and enforcement scenarios.

7.1 Device Selection Criteria

Devices selected for pilot must: - Be field-representative (e.g., laptop models used by engineers or operators) - Have TPM 2.0 and UEFI boot - Be capable of cellular (5G SIM) use - Not be in active production use during the pilot

7.2 Validation Matrix

Test Area	Expected Outcome	Notes
Hybrid Join Status	AzureAdJoined = YES; DomainJoined = YES	via dsregcmd /status
Compliance Result	Device shows as compliant in Intune	Intune Compliance blade
Offline Access Test	Cached login works; BitLocker + firewall enforced	With no network
Duo MFA Behaviors	MFA triggered on login and resource access	Simulate various login states
USB Tool Validation	Approved VID/PID devices work; all others blocked	via Defender logs / MDE portal

Each device must pass all five validation points before proceeding to full deployment.

7.3 Pilot Log Template

Create a log template for each test device:

Field	Entry Example
Device ID	NWR-PILOT-01
Serial Number	123456789XYZ
Test Performed By	Technician Name
Date of Testing	YYYY-MM-DD
Hybrid Join Status	AzureAdJoined: YES
Compliance Result	Compliant
BitLocker Status	Enabled; Key Escrowed
USB Device Check	Allowed: Tool1 VID 1234; Blocked: Storage VID 5678
Offline Login	Success with cached credentials
MFA Behavior	Prompted on Teams and VPN login
Screenshots Attached	Yes (saved to SharePoint folder)
Additional Notes	Device rebooted 2x to confirm persistence

7.4 Post-Pilot Actions

- Review all pilot logs and screenshots with internal IT and project sponsor
- Document any anomalies and prepare mitigation plans
- Approve rollout for production phase once pilot devices show 100% compliance

8. Third-Party Integration and Gaps

This section outlines the non-Microsoft tooling requirements and current technical gaps that need to be addressed through licensed third-party solutions or process changes.

8.1 DNS Filtering (Cisco Umbrella or Zscaler)

Use Case Dependency: Restricting 5G SIM internet access to M365/VPN-only.

Current Gap: Intune lacks native DNS content filtering per network interface (e.g., SIM vs. Wi-Fi) - Umbrella or Zscaler needed to enforce allowlist DNS rules

Recommendation: Procure Cisco Umbrella DNS Security Essentials licenses for all field devices - Integrate Umbrella agent deployment with Intune Win32 app package

8.2 USB Device Control (Defender E5 vs. Third-Party)

Use Case Dependency: USB-based IP tools and USB-to-serial whitelisting

Current Gap: Defender for Endpoint (E5) is required to apply VID/PID allow/block lists - If E5 is not licensed, a tool like DeviceLock or Safend must be used

Recommendation: If E5 not feasible, trial DeviceLock DLP and integrate with endpoint agent deployment - Build policy management for USB exception handling across field use cases9.4 Compliance Reporting & Audit Readiness

8.3 Patch Management (SICE Console)

Use Case Dependency: Remote patching of field devices over VPN

Current Gap: SICE agent must be validated in filtered network environment - Patch jobs may fail if DNS or VPN tunnel drops

Recommendation: Test failover logic in SICE agent - Explore fallback via Intune LOB update deployment or Winget automation

8.4 Offline Content & Policy Enforcement

Use Case Dependency: Secure offline use in no-network zones

Current Gap: Some Intune policies do not persist without internet refresh - Defender signatures may go stale if disconnected >30 days

Recommendation: Explore use of Windows Update for Business cache or on-prem WU caching - Create SOP for re-connection checkpoints every 14–21 days

8.5 Summary Table

Dependency Area	Third-Party Required	Current Status	Mitigation Plan
DNS Filtering	Cisco Umbrella	Not licensed	Purchase Essentials Tier, deploy agent
USB Control	DeviceLock (optional)	Not deployed	Evaluate based on Defender E5 status
Patch Management	SICE MSP Console	Agent under test	Add fallback via Intune or Winget
Offline Policy Sync	None	Known issue	Schedule reconnect intervals

9. Security and Compliance Controls

This section outlines how Microsoft Intune, Defender, and Azure AD are used to enforce security posture and meet compliance expectations for field device deployments. Controls align with standard frameworks (e.g., CIS, NIST) and support regulatory readiness.

9.1 Security Objectives

- Ensure only compliant devices can access corporate resources
- Maintain endpoint encryption, antivirus, and firewall posture
- Restrict use of unauthorized apps and peripherals (e.g., USB)
- Enforce MFA and Conditional Access for identity protection

9.2 Intune Enforcement Areas

Category	Policy Type	Description
Device Encryption	Compliance + BitLocker	Devices must have TPM-backed BitLocker enabled
Antivirus	Compliance + Baseline	Defender AV must be running and up to date
Firewall	Baseline + Settings Catalog	Firewall must be active on all profiles
USB Controls	Defender (E5) / DeviceLock	Block unapproved USB tools, allow exceptions
Local Admin Block	Config Profile	Remove end-user from local Administrators group
App Restrictions	Intune App Protection	Optional: Restrict LOB/3rd party app installs

9.3 Identity and Access Controls

Control Layer	Enforcement Tool	Description
MFA Enforcement	Azure AD / Duo	All users prompted for MFA during login and resource access
Conditional Access	Azure AD CA	Blocks access from non-compliant, unregistered, or risky devices
Hybrid Join Trust	GPO + Entra ID	Validates hybrid-joined status before applying MDM profiles

Additional Measures: Risk-based conditional access: Trigger additional prompts for risky sign-ins - User sign-in reports for anomaly detection

9.4 Compliance Reporting & Audit Readiness

- Use Microsoft Secure Score to measure improvement
- Configure Microsoft Purview (if available) for DLP and audit logging
- Export compliance status via Intune Reports > Devices > Noncompliant Devices
- Periodic compliance snapshot exported monthly to SharePoint or PDF

Retention: Audit logs retained 180 days (default) - Compliance reports stored for a minimum of 12 months for audit purposes

9.5 Compliance Reporting & Audit Readiness

Framework	Control Mapped	Source Tool
CIS v8	3.4, 4.1, 5.2, 6.5, 13.1	Intune, Defender, Entra ID
NIST 800-53	AC-3, AC-7, IA-2, SC-28, CM-7	Azure AD, Intune
ISO 27001	A.6.2.1, A.9.2.3, A.12.6.2	Compliance Portal, CA
Essential Eight	Application Control, Patch Applications, Config Hardening, MFA, Admin Restrictions	Intune, Defender for Endpoint, Entra ID
IEC 62443	SR 1.2, SR 1.3, SR 2.1, SR 4.1, SR 6.2	Intune, Defender for IoT, Azure AD

10. SOPs & Admin Runbooks

This section documents operational procedures and runbooks for administrators, support teams, and field engineers. These SOPs ensure consistent and auditable handling of endpoint scenarios from deployment to offboarding.

10.1 SOP-001: New Device Enrollment

Audience: IT Helpdesk, Field Support

Steps: 1. Boot device into Windows Autopilot or use provisioning package 2. Connect to corporate Wi-Fi or 5G SIM 3. Authenticate with corporate account 4. Device auto-enrolls into Intune and applies policies 5. Validate compliance in Endpoint Manager

Checklist: Device joined, policies applied, BitLocker enabled, Defender active

10.2 SOP-002: Device Offboarding and Decommissioning

Audience: IT Admin / Security

Steps: 1. Revoke user access in Azure AD 2. Remove device from Intune using “Retire” or “Wipe” action 3. Confirm BitLocker recovery key and device log archival 4. Physically decommission device or reassign

Security Note: Devices must be wiped using Intune to remove MDM profile and corporate data

10.3 SOP-003: USB Device Approval Workflow

Audience: Security Admin

Steps: 1. User submits USB VID/PID for business justification 2. Security team validates use case 3. Add device ID to Defender allow list (or DeviceLock rule) 4. Push updated policy via Intune

Audit Log: Maintain USB allow-list history in SharePoint or SIEM

Additional Measures: Risk-based conditional access: Trigger additional prompts for risky sign-ins - User sign-in reports for anomaly detectionUse

10.4 SOP-004: Troubleshooting Hybrid Join & Intune Enrollment

Audience: Field Tech / Tier 2 IT Support

Checklist: Run dsregcmd /status - Confirm SCP GPO is linked - Review event logs (Event ID 304 / 1098) - Verify MDM enrollment registry path - Test user login + Conditional Access result

Tooling: Event Viewer, Regedit, Intune Portal, PowerShell

10.5 SOP-005: Offline Login and Field Device Recovery

Audience: Field Engineer / Onsite Support

Steps: 1. Attempt cached credential login 2. Check local BitLocker and Defender status 3. Log findings in offline template form 4. Contact central IT if MFA block persists beyond 72 hours

Contingency: Field USB tool can be preapproved for unlocking kiosk/admin shell if urgent

10.6 SOP-006: Monthly Compliance Export and Review

Audience: Compliance Officer / IT Manager

Steps: 1. Navigate to Intune > Devices > Monitor > Compliance 2. Export non-compliant device list 3. Store export on SharePoint ComplianceReports/YYYY-MM 4. Summarize trends and gaps in Power BI or Excel

Frequency: Monthly or as required by audit schedule

11. Assumptions & Constraints

This section outlines the foundational assumptions and known constraints used to define the project scope, technical feasibility, and operational readiness for Intune + Windows Device Management deployment.

11.1 Assumptions

Category	Assumption
OS Platform	All laptops are running Windows 10/11 Pro or Enterprise
Device Capability	All devices have TPM 2.0, UEFI, and SIM support (where needed)
Licensing	Intune, Azure AD Premium P1, and Defender Plan 2 licenses are available
Admin Access	Global Admin rights are available for Intune and Azure configuration
Network Access	Devices have outbound internet access (or approved proxy for Intune reach)
Third-Party Software	Cisco Umbrella, DeviceLock, and SICE patch console licenses can be procured
Field Access	Physical access is available for initial device provisioning and testing

11.2 Constraints

Area	Constraint
DNS Filtering	No native SIM-based DNS enforcement in Intune (requires Umbrella/Zscaler)
USB Control	Defender for Endpoint E5 is required for granular VID/PID control
Offline Support	Intune compliance check-in may be delayed in no-network environments
Driver Enforcement	Some USB-to-serial drivers may require manual intervention or staging
SICE Agent Integration	Patch agent may not support full headless deployments on all device models
Policy Refresh	Cached policies may expire if device is offline >30 days
Language/Locale	All configurations and UI expected in English (multi-language support not scoped)

Note: Any deviation from these assumptions may require revisiting design decisions or planning for exceptions in future iterations.

12. Appendices

This section provides raw artifacts, evidence, and backup data to support configuration, testing, and operational readiness. All files should be retained and versioned as part of the delivery package.

12.2 Policy Exports (JSON/XML)

Policy Area	Format	File Name
Compliance Policy	JSON	BitLocker-CompliancePolicy.json
Config Profile	XML	WiFi-VPN-Kiosk-Profile.xml
Security Baseline	JSON	Microsoft-SecurityBaseline.json
Group Assignments	CSV	Intune-Group-Mapping.csv

12.4 Pilot Logs and Validation Reports

File	Location	Format
Pilot Log Template	/PilotLogs/Template.xlsx	Excel
Pilot Device Results	/PilotLogs/Results/	PDF
USB Test Logs	/PilotLogs/USB/	TXT

12.5 Reference Documents and Links

- Microsoft Hybrid Join: https://aka.ms/HybridJoin
- Intune Compliance: https://aka.ms/IntuneCompliance
- Cisco Umbrella DNS: https://umbrella.cisco.com/
- Defender USB Control: https://learn.microsoft.com/en-us/microsoft-365/security/defender-endpoint/device-control

13. Change Log & Document History

This section provides a record of version changes and updates made to this implementation document throughout its lifecycle. It is intended to support traceability and accountability across stakeholders.

13.1 Version History Table

Version	Date	Author	Description of Changes
1.0	25 June 2025	Vipin Dubey	Initial release for review
1.1	TBD	TBD	Future revision - feedback implementation

15. Risk Management & Mitigation Strategy

This section outlines identified risks across the implementation phases and provides a mitigation plan to ensure minimal business disruption, data protection, and operational continuity. Each risk is categorized by impact level, with mitigation strategies, escalation paths, and rollback plans clearly defined.

15.1 Risk Table Overview

Risk Description	Impact	Mitigation Measures	Escalation Contact/Team	Rollback Plan
Wi-Fi disablement not persistent	Medium	Use recurring PowerShell task and enforce GPO restrictions	Field Support Lead	Re-enable adapter via signed PowerShell recovery script
MFA misconfiguration blocks access	High	Test policies in pilot, use break-glass accounts, validate with CA logs	Azure Security Admin	Disable CA policy temporarily, switch to cached credential use
Device fails compliance post-enroll	Medium	Use phased policy assignment, monitor with compliance reports	Intune Admin	Move device to quarantine group, reapply staged profiles
USB tool blocked	Medium	Validate VID/PID, run Defender logs before rollout	Endpoint Security Engineer	Remove conflicting Defender policy, reapply updated whitelist

despite whitelist				
Offline login fails	Medium	Enable credential caching via GPO, pre-validate on test devices	Desktop Engineering	Connect to trusted network, re-sync GPO and MDM policies
BitLocker escrow fails	High	Verify Azure AD join status before encryption begins	Intune + Security Team	Recover manually, re-encrypt after backup key validation

16. Intune Admin Roles & Access Control Design

This section defines the role-based access control (RBAC) structure for Microsoft Intune and Azure AD (Entra ID). It ensures responsibilities are delegated securely and aligns with the principle of least privilege.

16.1 Role-Based Access Model (RBAC)

Role Name	Description	Scope	Assigned Team
Intune Administrator	Full access to create, modify, assign all policies	Global	Lead Admin
Policy Author	Create/edit policies but not assign	Targeted groups	Endpoint Engineers
Helpdesk Operator	Perform remote actions (wipe, reset, locate)	Devices group only	Tier 1 IT Support
Security Reader	View compliance, CA, audit logs	Read-only global	Security/Compliance Team
Enrollment Manager	Manage Autopilot profiles and device registration	Autopilot device objects	Provisioning Technician

RBAC is enforced in both Intune and Entra ID. Assignment is scoped to groups and devices using least-privilege delegation.

16.2 Delegation & Role Assignment

- **Intune roles** are configured via Endpoint Manager Admin Center > Tenant Administration > Roles
- **Azure AD roles** are managed in Entra ID and optionally enforced using **PIM (Privileged Identity Management)** for JIT activation
- All administrative accounts must be protected by **MFA** and assigned per SOP-006

16.3 Audit Logging Configuration

- Enable **Audit Logs** in Entra ID: Logs role changes, group updates, sign-in behavior
- Enable **Device compliance audit** in Intune Reports > Audit Logs
- Store reports to central SharePoint: /AuditLogs/YYYY-MM
- Recommended retention: **180 days minimum**

16.4 Least Privilege Enforcement Principles

- No permanent Global Admin or Intune Admin roles
- Break-glass accounts used only in emergency
- All device-specific actions scoped to device groups
- Role permissions reviewed quarterly (see Section 18: Governance)

Delegation Notes: Use Azure AD PIM to enforce just-in-time role activation.

Audit Logging:

- Enable unified audit log in Microsoft Purview
- Enable Intune change tracking (policy edits, enrollments)

17. Reporting and Monitoring Strategy

This section outlines how Intune, Azure AD, and Defender reporting features are used to monitor compliance, generate actionable insights, and ensure proactive enforcement. It includes structured templates, alert configurations, and export paths.

17.1 Reporting Templates

Report Name	Frequency	Purpose	Format / Tool
Device Compliance Snapshot	Monthly	Overall compliance per policy category	CSV / Intune GUI
Noncompliant Devices Trend	Weekly	Track recurring policy failures	Power BI / Excel
Audit Log Summary	Monthly	Track changes to roles, policies, enrollments	CSV / Entra ID
Enrollment Failures	Ad hoc	Troubleshoot onboarding issues	JSON / Graph API

17.2 Data Export Options

- **Intune Admin Center:**
  - Devices > Monitor > Reports > Export CSV
- **Graph API:**
  - /deviceManagement/reports/exportJobs
  - Authentication via Azure App Reg + delegated permissions
- **SharePoint Folder Structure:**
  - /Reports/YYYY-MM/Compliance.csv
  - /Reports/YYYY-MM/AuditLogs.csv

17.3 Dashboards and Visualization

- **Power BI Integration:**
  - Use Intune Data Warehouse or CSV connector
  - Build dashboards for:
    - Device compliance by region
    - Weekly compliance drift
    - Top policy failures by type
- **Excel Templates:**
  - Pivot tables for quick compliance snapshots
  - Heatmaps for missed patch trends

17.4 Alerting and Monitoring

Alert Type	Trigger Condition	Configured In
Compliance Deviation	>5% of devices noncompliant	Intune / Power Automate
Role Assignment Change	Any admin role added	Entra ID audit alerts
USB Policy Violation	Unauthorized USB connection detected	Defender for Endpoint
MFA Failure Spike	>10% MFA failure rate in 1 hour	Entra ID Sign-In Logs

Notification Channels: Email, Teams Webhook, or SIEM alerts

18. Operational Governance Framework

Device Management environment. It includes decision-making policies, change controls, and user engagement procedures.

18.1 Device Lifecycle Governance

Lifecycle Stage	Action Owner	SOP Reference	Notes
New Deployment	Field Support Team	SOP-001	Device added via Autopilot or manual config
Device Retirement	IT Admin	SOP-002	Use Intune retire/wipe + asset removal
Break/Fix Handling	Helpdesk / Tier 2	SOP-004	Re-enroll or restore device as needed

18.2 Change Control Process

Policy Change Workflow:

1. Change request logged in SharePoint or ITSM system
2. Review by Policy Committee (monthly or urgent basis)
3. Pilot applied to test group (minimum 2 devices)
4. Feedback reviewed → Final approval
5. Deployment to production groups

Scope Includes:

- Compliance policy updates
- Configuration profile changes
- Baseline version upgrades

18.3 Monthly Policy Review Checklist

18.4 End-User Feedback and Communication

Channels:

- Email feedback form distributed quarterly
- Self-service Intune support page (FAQ + ticket link)
- Targeted feedback via helpdesk follow-ups

Goal: To identify usability issues, blocked apps or hardware, and process friction before they escalate.

18.5 Patch and Update Governance

Component	Frequency	Managed By	Notes
OS Updates (WUfB)	Monthly	Endpoint Team	Enforced via Update Rings
Defender Signatures	Daily	Security / Defender	Synced via cloud + offline fallback
SICE Patch Jobs	Monthly	MSP / Field IT	Validated post-deployment by reports

19. Automation & Future Enhancements

This section highlights automation opportunities using Microsoft Graph, Azure Automation, Defender integrations, and CI/CD practices to enhance scalability and operational efficiency.

19.1 Graph API for Bulk Operations

Task	API Endpoint Example	Notes
Export compliance reports	/deviceManagement/reports	Schedule via Logic App
Assign config profiles to group	/deviceManagement/deviceConfigurations/{id}/assign	Automate policy deployment
Create dynamic groups	/groups with rule syntax	Useful for USB exemptions or pilots
Retire/wipe devices	/managedDevices/{id}/retire	Enable auto-cleanup or quarantine

19.2 Azure Automation Opportunities

- Automate cleanup of noncompliant devices >30 days
- Trigger re-enrollment for devices with failed policy sync
- Scheduled tasks to pull compliance summary and upload to SharePoint

Tools:

- Azure Automation Account
- PowerShell Runbooks
- Managed Identity for secure API access

19.3 Defender for Endpoint Enhancements

- Use attack surface reduction telemetry for USB policy validation
- Integrate Defender alerts with Microsoft Sentinel or email workflows
- Enable device tagging based on risk score or compliance state

19.4 CI/CD for Intune Config Management

- Store policy JSONs and scripts in GitHub/GitLab repo
- Use GitHub Actions or Azure DevOps pipeline to deploy to test tenants
- Automate change log generation and rollback version control

20. Glossary & Acronym Reference (1 page)

This glossary provides definitions for key acronyms and technical terms used throughout the Intune + Windows Device Management implementation plan. It ensures clarity and consistency across stakeholders, including IT admins, auditors, and business sponsors.

Term / Acronym	Full Name / Definition
AAD / Entra ID	Azure Active Directory, now known as Microsoft Entra ID
Intune	Microsoft Intune - Mobile Device Management (MDM) and endpoint policy platform
MDM	Mobile Device Management - for deploying and enforcing device-level policies
MAM	Mobile Application Management - controls app access and usage
ASR	Attack Surface Reduction - Defender capability to reduce exploitable paths
SCP	Service Connection Point - AD object used for Hybrid Azure AD Join
PIM	Privileged Identity Management - for just-in-time access control
RBAC	Role-Based Access Control - permission management model used in Azure and Intune
GPO	Group Policy Object - legacy configuration tool for Windows environments
CSP	Configuration Service Provider - used to deliver Intune settings to Windows devices
CA	Conditional Access - controls access to apps based on compliance, identity, or context
E5	Microsoft 365/Defender Plan 2 license tier with advanced security capabilities
WUfB	Windows Update for Business - policy-based update scheduling and deferral
LOB App	Line-of-Business Application - internal or company-specific application

21. Licensing Requirements Breakdown

This section outlines the licensing components required to implement and sustain the Intune + Windows Device Management solution as described in this document. It also provides a cost estimate for pilot and production rollouts.

21.1 License Requirements Matrix

License Type	Purpose	Used For	Notes
Microsoft Intune (EMS E3)	Core device management & MDM	Enrollment, compliance, config policies	Required baseline license
Azure AD Premium P1	Identity & hybrid join support	Conditional Access, Hybrid Join, Dynamic Groups	Included with EMS E3
Microsoft Defender for Endpoint Plan 2	Endpoint protection & USB control	ASR rules, device tagging, USB enforcement	Optional, adds high value for security
Azure AD Premium P2	Advanced identity protection	PIM, risky sign-ins, identity governance	Optional, not required for all users
Microsoft 365 E3/E5	User productivity & licensing bundle	Covers Office, Exchange, Teams, plus EMS	Only if bundled licensing is preferred

21.2 Estimated Licensing Cost (Monthly)

User Count	License Bundle	Estimated Monthly Cost (USD)	Notes
10 (Pilot)	EMS E3 + Defender	$220	10 users $22/user
100 (Prod)	EMS E3 + Defender	$2,200	Assumes Defender Plan 2 is included
100 (Alt)	Intune Only (E3)	$1,600	Without Defender or P2 functionality

Cost Planning (Estimated):

- Pilot (10 users): $220/month
- Full rollout (100 users): $2,200/month

22. Backup and Recovery Strategy

This section defines the backup practices and recovery actions required to protect configuration integrity, ensure business continuity, and streamline device restoration for the Intune and Windows Device Management environment.

22.1 Policy and Configuration Backup

Artifact	Backup Format	Backup Location	Frequency
Intune policies & profiles	JSON/XML	SharePoint / Git repo	Monthly
GPO exports	GPO Backup Set	On-prem AD backup folder	After each edit
Pilot test scripts	.ps1 / .bat	/Scripts in repo	Static versioned
Device compliance snapshots	CSV	/Reports/YYYY-MM/Compliance.csv	Weekly

22.2 BitLocker Key Handling

- Keys stored automatically in Azure AD (Entra ID)
- Validate escrow after encryption via Get-BitLockerRecoveryPassword
- Periodically audit recovery keys for availability
- Export list of recovery key status monthly for audit readiness

22.3 Lost/Stolen Device Recovery Process

Step	Action	Tool Used
1	Mark device as lost	Intune Portal
2	Issue remote lock or wipe	Intune > Devices
3	Notify Security / Compliance team	Teams + Email Alert
4	Revoke user tokens and sessions	Entra ID Sign-In Logs
5	Replace device and re-enroll via SOP-001	Field Support Team

22.4 Recovery Planning

- Maintain signed PowerShell scripts for re-enrollment and profile repair
- Snapshot baseline policy folder after each major update
- Store change log in /ChangeHistory.md
- Include rollback policies in /Recovery folder of repo

23. Key Performance Indicators (KPIs) and SLAs

This section defines the critical success metrics that will be used to track and measure implementation effectiveness and operational performance. Each KPI is mapped to a tool or method of measurement, with suggested response timelines and SLA thresholds.

23.1 Implementation Phase KPIs

KPI	Target Value	Description	Monitoring Method
BitLocker Compliance Rate	100% within 5 days	Devices must show encryption with key escrow	Intune Compliance Dashboard
Wi-Fi Bypass Rate	<5%	Measure if Wi-Fi disablement policy was circumvented	Defender Alerts, Field Reports
Pilot Enrollment Time	<48 hours downtime	Time taken to onboard all pilot devices	Autopilot + Enrollment Logs
Hybrid Join Success Rate	>98%	Devices should show Azure AD + Domain Join	dsregcmd, AAD Portal
Policy Enforcement Time	<30 minutes	Time from enrollment to policy compliance	Intune Timeline Reports

23.2 Operations & Support KPIs

KPI	Target Value	Description	Monitoring Method
MFA Login Success Rate	>98%	Users successfully authenticate with MFA	Entra ID Sign-In Logs
Helpdesk Ticket Acknowledgement Time	<4 hours	SLA for Tier 1 support ticket acknowledgment	ITSM System
Monthly Compliance Drift	<10%	Percentage of devices drifting from compliance	Power BI Trend Analysis

Patch Deployment Completion	>95% within 7 days	Timely OS and Defender patching	SICE + Intune Reports

23.3 Monitoring Plan

- **Daily Dashboards:** Power BI visual tracking of top-line KPIs
- **Weekly Reviews:** Intune snapshot reports and drift analysis
- **Monthly Reports:** Formal audit and SLA compliance overview
- **Escalation Path:** Breach of any SLA triggers Incident Manager review

Each metric will be reviewed by the Security Governance team monthly and refined as rollout scales.